

## Apostol, Chapter 1 - Exercise 16

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**16. Prove that if  $2^n - 1$  is prime, then  $n$  is prime.**

Suppose the  $n$  is composite, that is,  $n = ab$ , for some integers  $a, b > 1$ .  
For any  $j > 1$ , we have that

$$x^j - 1 = (x - 1)(x^{j-1} + x^{j-2} + \dots + x + 1).$$

Taking  $x = 2^a$  and  $j = b$  we get

$$2^n - 1 = 2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \dots + 2^a + 1)$$

Since  $a > 1$ , we have that  $2^a - 1 > 1$  and  $2^{a(b-1)} + 2^{a(b-2)} + \dots + 2^a + 1$ .  
Thus,  $2^n - 1$  is not a prime (Contradiction).  
Therefore,  $n$  is prime.